GUAIFENESIN

Chesty cough
Drink fluids

CHEMICAL NAME (RS)-3-(2-methoxyphenoxy)propane-1,2-diol (glyceryl guaiacolate)
CLASS Expectorant
TRADE NAMES Robitussin · Mucinex · Benylin Chesty Cough · Tussin

THERAPEUTIC USES / INDICATIONS

- Productive (chesty) cough — to loosen and thin mucus.
- Common cold and acute bronchitis (symptomatic).

ADMINISTRATION

- **PO** syrup or tablets with a full glass of water.
- ER tablets swallowed whole.

DOSAGES

- **Adult:** 200–400 mg every 4 h; max 2.4 g/day (ER 600–1,200 mg every 12 h).
- **Children 6–12 yr:** 100–200 mg every 4 h (max 1.2 g/day).
- Not recommended <6 yr.

PHARMACOKINETICS

- **Onset:** 30 min. **Duration:** 4–6 h.
- Metabolised in liver.
- Excreted in urine. **t½:** 1 h.

PHARMACODYNAMICS

- Increases volume and reduces viscosity of respiratory secretions (via gastric vagal reflex) → easier expectoration.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting (high doses).
- Dizziness, headache.
- Rash.

ADVERSE EFFECTS (SERIOUS)

- Kidney stones with chronic high-dose misuse (rare).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- Children <6 yr (OTC).

KEY INTERACTIONS

- Combination products may contain antihistamines, decongestants or dextromethorphan; check.

NURSING MANAGEMENT

- Encourage fluids and deep breathing/coughing.
- Assess sputum colour and amount.
- Cough >3 weeks: investigate.

PATIENT EDUCATION

- Drink plenty of water to help loosen mucus.
- Read labels of combination products.
- See a doctor if cough persists, fever is high or sputum is bloody.

KEY POINTS

- Expectorant; thins mucus.
- Drink plenty of fluids.
- For productive cough.
- Max 2.4 g/day.

SOURCES CHECKED **[477]** 21 CFR Part 341 (US OTC monograph) **[495]** Guaifenesin Extended Release 600 mg, oral (US label)
Full references p. 432–457